

Benchmarking Local Large Language Models for Structured Knowledge Extraction in Conversational Commerce

Ratomir Karlović ¹ , Luka Sever ¹ , Sandi Baressi Šegota ¹ , Vedran Mrzljak ²  and Ivan Lorencin ¹ 

¹ Faculty of Informatics, Juraj Dobrila University of Pula, Zagrebačka 30, 52100 Pula, Croatia; ratomir.karlovic@unipu.hr (R.K.), luka.sever@unipu.hr (L.S.), sandi.baressi.segota@unipu.hr (S.B.Š.), ivan.lorencin@unipu.hr (I.L.)

² Faculty of Engineering Rijeka, University of Rijeka, Vukovarska 58, 51000 Rijeka, Croatia;

* Correspondence: vedran.mrzljak@riteh.uniri.hr;

Abstract

The integration of natural language understanding into conversational commerce requires robustly extracting structured intents from idiomatic text. While cloud-based models are highly capable, their latency, privacy risks, and operational costs necessitate efficient local alternatives. To address this, we empirically evaluate locally deployed, open-weights large language models (LLMs) on parsing multi-intent shopping cart commands into deterministic JSON schemas. We benchmark multiple models across minimal, extended, and few-shot prompting paradigms, utilizing a dual-axis framework to simultaneously measure extraction quality (operation F1-score, exact match, schema validity) and hardware performance (latency, GPU power consumption, VRAM usage). Our evaluation highlights a fundamental trade-off between schema compliance and computational overhead, demonstrating how advanced prompting techniques improve parsing accuracy while measurably impacting inference latency and power draw. Ultimately, this comprehensive profiling identifies optimal model configurations for resource-constrained environments, establishing a practical baseline for deploying cost-effective and energy-efficient LLMs in edge-based retail automation.

Keywords: LLM; Prompt Engineering; Natural Language Understanding; JSON Parsing; Structured Output; Model Performance; Conversational Agents; Information Extraction;

1. Introduction

The template details the sections that can be used in a manuscript. Note that the order and names of article sections may differ from the requirements of the journal (e.g., the positioning of the Materials and Methods section). Please check the instructions on the authors' page of the journal to verify the correct order and names. For any questions, please contact the editorial office of the journal or support@mdpi.com. For LaTeX-related questions please contact latex@mdpi.com.

2. Background and Related Work

3. Methodology

Materials and Methods should be described with sufficient details to allow others to replicate and build on published results. Please note that publication of your manuscript implicates that you must make all materials, data, computer code, and protocols associated with the publication available to readers. Please disclose at the submission stage any

Received:

Revised:

Accepted:

Published:

Copyright: © 2026 by the authors.

Submitted to *Journal Not Specified* for possible open access publication under the terms and conditions of the

[Creative Commons Attribution \(CC BY\)](#) license.

restrictions on the availability of materials or information. New methods and protocols should be described in detail while well-established methods can be briefly described and appropriately cited.

Research manuscripts reporting large datasets that are deposited in a publicly available database should specify where the data have been deposited and provide the relevant accession numbers. If the accession numbers have not yet been obtained at the time of submission, please state that they will be provided during review. They must be provided prior to publication.

Interventionary studies involving animals or humans, and other studies require ethical approval must list the authority that provided approval and the corresponding ethical approval code.

In this section, where applicable, authors are required to disclose details of how generative artificial intelligence (GenAI) has been used in this paper (e.g., to generate text, data, or graphics, or to assist in study design, data collection, analysis, or interpretation). The use of GenAI for superficial text editing (e.g., grammar, spelling, punctuation, and formatting) does not need to be declared.

4. Results and discussion

5. Limitations and Future Work

6. Conclusions

This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used “Conceptualization, X.X. and Y.Y.; methodology, X.X.; software, X.X.; validation, X.X., Y.Y. and Z.Z.; formal analysis, X.X.; investigation, X.X.; resources, X.X.; data curation, X.X.; writing—original draft preparation, X.X.; writing—review and editing, X.X.; visualization, X.X.; supervision, X.X.; project administration, X.X.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript.”, please turn to the [CRediT taxonomy](#) for the term explanation. Authorship must be limited to those who have contributed substantially to the work reported.

Funding: This work was (partially) supported by the EC Digital Europe Programme EDIH Adria 2.0 (101256325); SPIN projects IP.1.1.03.0120, IP.1.1.03.0158 and IP.1.1.03.0039; and NextGenerationEU University grants: uniri-iz-25-6, uniri-iz-25-220, IIP_010144, IIP_010136

Data Availability Statement: The original data presented in this study are openly available in a public repository at <https://www.mdpi.com/ethics>.

Acknowledgments: This research was performed using the Advanced computing service provided by University of Zagreb University Computing Centre - SRCE.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

LLMs	Large Language Models
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
IA ³	Infused Adapter by Inhibiting and Amplifying Inner Activations
ICL	In-Context Learning
JSON	JavaScript Object Notation
SFT	Supervised Fine-Tuning
F1	F1 Score
GPU	Graphics Processing Unit
DDP	Distributed Data Parallel
MLP	Multi-Layer Perceptron
bf16	BFloat16 Precision Format
tf32	TensorFloat-32

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.